MILITARY SPECIFICATION SHEET

BATTERY, STORAGE, AIRCRAFT, NICKEL-CADMIUM, 24 VOLTS, 20 AMPERE-HOUR (BB-434/A)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the battery assembly described herein shall consist of this document and the latest issue of specification MIL-B-81757.

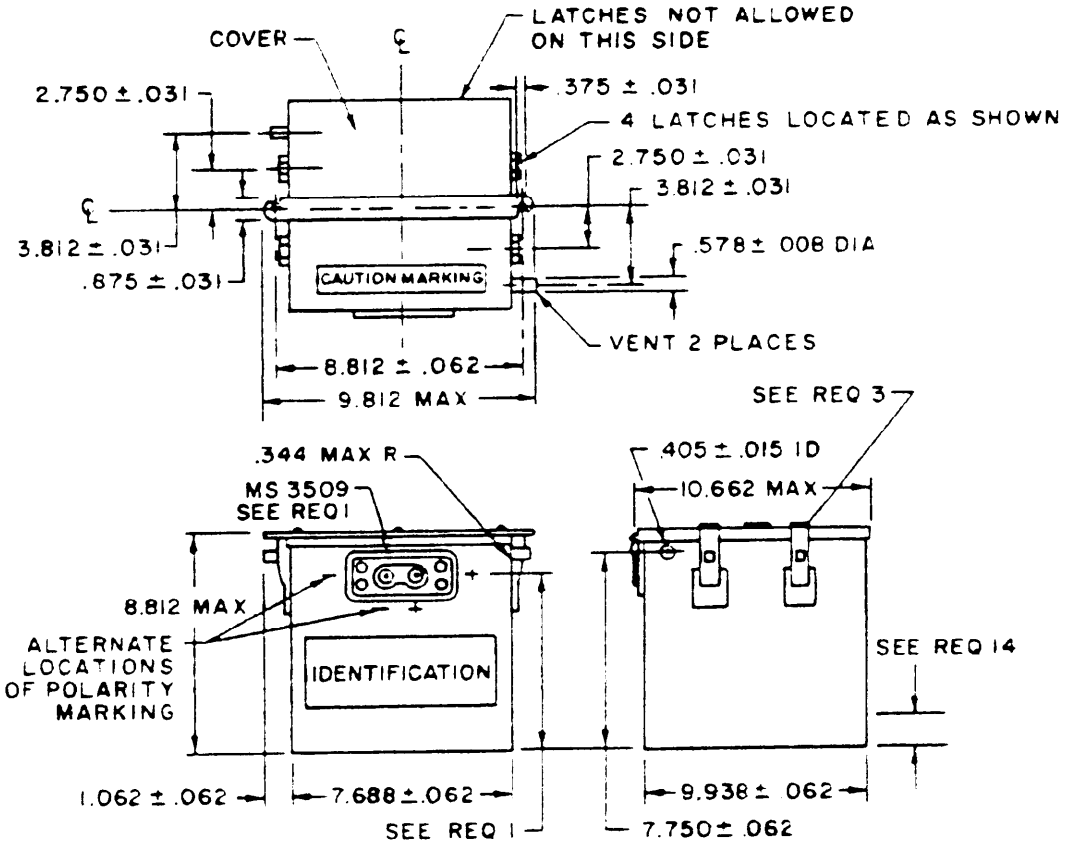


Figure 1. Battery assembly.
P/N M81757/8-4 and M81757/8-5

(D)

FSC 6140

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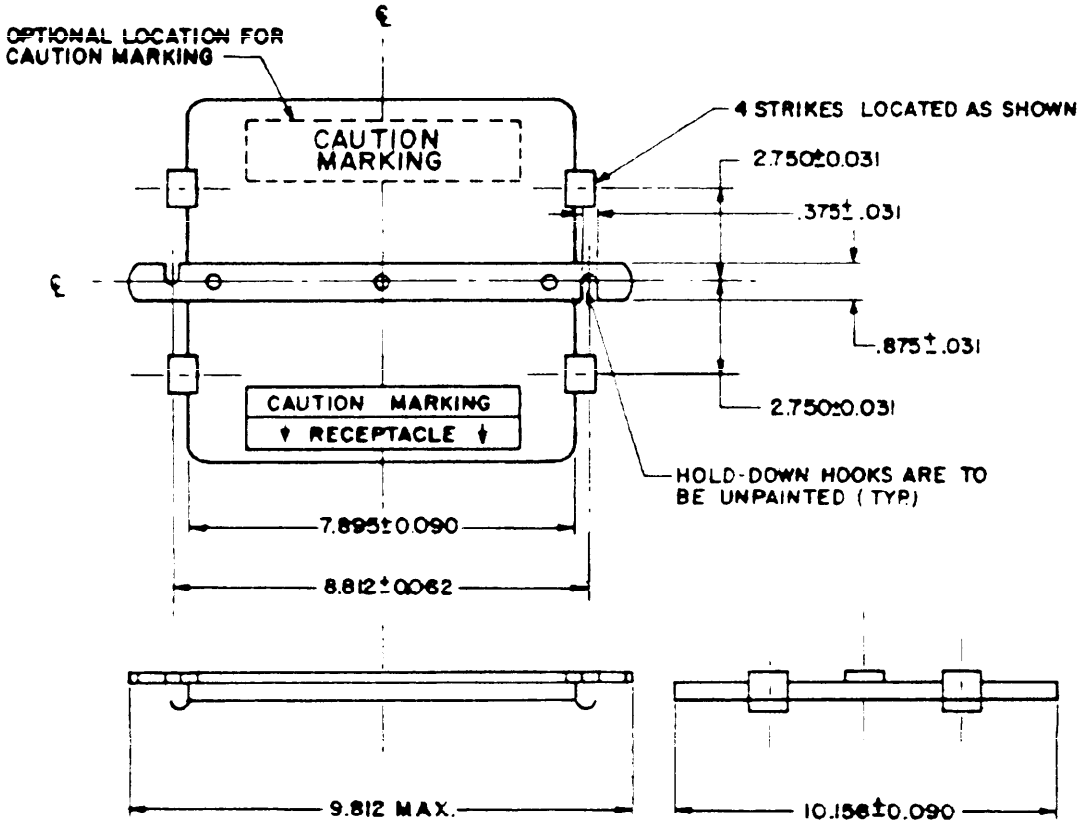


Figure 2. Cover.

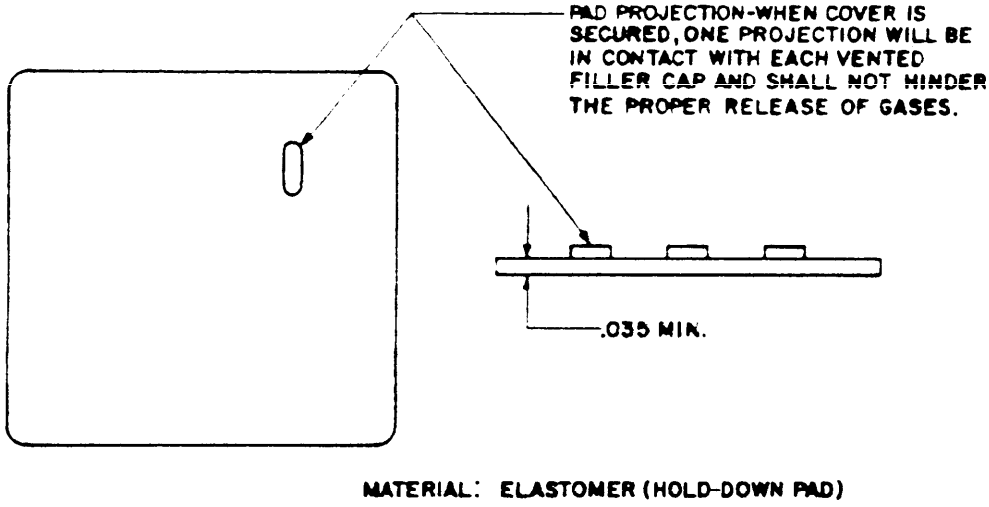


Figure 3. Cover hold-down pad.

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CONNECTOR STRAP ARRANGEMENT
(D) BATTERY ASSEMBLY P/N M81757/8-4, /8-5

REFERENCE LETTER	STRAP PART NO.	DESCRIPTION	QTY
A	M81757/3-5	Connector, Strap	2
B	M81757/3-6	Connector, Strap	3
C	M81757/3-7	Connector, Strap	10
E	M81757/5-1	Connector, Curved	1
F	M81757/5-2	Connector, Curved	1
G	M81757/3-9	Connector, Strap	1
H	M81757/4-3	Connector, Tab	2

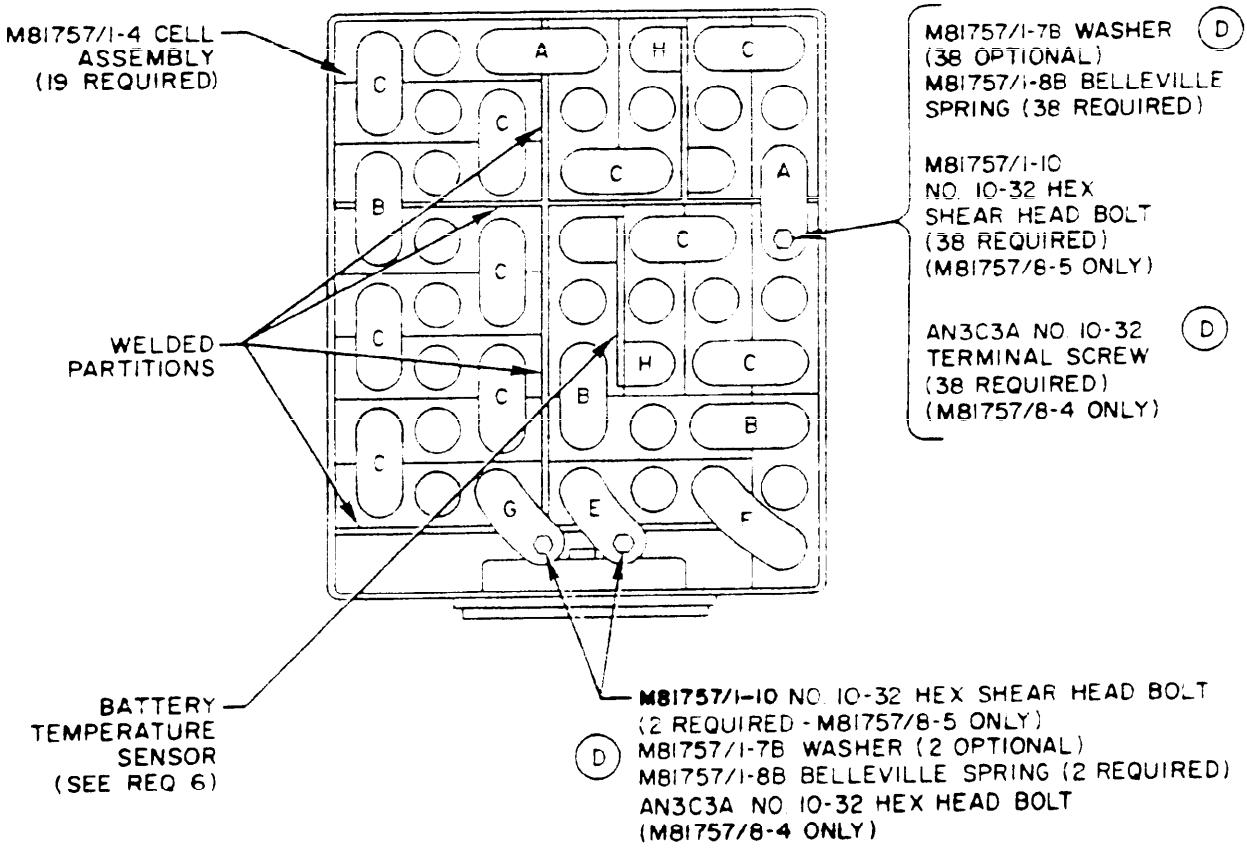


Figure 4. Cell arrangement.

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REQUIREMENTS:

1. The receptacle terminal block shall be 0.866 ± 0.015 from the centerline of the contact pins to the top surface of the terminal block. The receptacle will be so located that the top of the terminal block and the cell terminal contact surfaces are in the same plane. Receptacle terminal block shall be screw size 10-32. Receptacle shall conform to MS3509-28.

2. All dimensions are prior to epoxy coating except: Overall length - 9.998 max., overall width - 10.662 max., and overall height - 8.812 max.

3. Method of mounting latches and strikes shall be optional. They shall conform to Corbin Cabinet Lock Division, American Hardware, Item Number 15834, or Nielson Latch Number C-833314-2-SS-302.

4. Spacers if used shall be polyethylene sheet per ASTM STD D4066-82, Comp. A, Type I and Type III, minimum 0.015 inch thick, color natural.

5. Latches and strikes shall not be epoxy coated.

6. A sensor (thermocouple or thermistor), when required shall be located as shown for the purpose of monitoring battery temperature during testing to this specification. The sensor shall be positioned in the approximate center of the cell container's widest outer surface (equidistant from the top and bottom and from side to side).

7. The battery voltage shall be 24 volts (nominal).

8. The battery weight shall be 55 pounds maximum when filled.

9. The battery shall contain 19 cells.

10. The rated capacity at 1C rate shall be 20 A.H./1 Hr/80°F/18.0V.

11. Length and width of cover pad and spacer shall conform to the inside dimensions of the applicable cover.

12. The length, width, and thickness of cover pad projections shall provide sufficient pressure on the vented filler caps to hold down the battery cells during vibration.

13. The quantity and location of cover projection pads shall be determined by the vented filler caps.

14. The base dimensions of the container shall remain within tolerance for a distance of 1.8 inches up from the bottom.

(D) 15. The M81757/8-4 battery is not for Navy use.

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D 16. The battery identification on the M81757/8-4 and M81757/8-5 battery shall indicate the correct supersession data as indicated in Note 2.

D 17. The M81757/8-5 battery shall incorporate M81757/1-10 bolts to secure the cell interconnecting strap, thereby preventing cell removal.

D 18. The "CAUTION" label on the M81757/8-5 battery shall contain the following additional information: "The cells in this battery are not to be removed."

19. The cell vented filler cap shall be in accordance with MIL-B-81757/2.

NOTES:

1. All dimensions are in inches, unless otherwise specified.

D 2. Supersession data: This specification sheet supersedes MS18045, MS90321, MS90365, MS24497 and MIL-B-81757 part numbers as follows, when a product has been qualified for each MIL-B-81757 part number respectively.

Part Number	Replaces Part Numbers
M81757/8-4	MS24497-3, MS24497-5, M81757/8-2
M81757/8-5 (for Navy use only)	M81757/8-2, MS18045-45, MS18045-49, MS90365-1, MS90365-2, MS90321-68W, MS90321-77, MS90321-78W, M81757/8-3

Part Number	Superseded By
MS18045-45	M81757/8-5
MS18045-49	M81757/8-5
MS24497-3	M81757/8-4
MS24497-5	M81757/8-4
MS90321-68W	M81757/8-5
MS90321-77W	M81757/8-5
MS90321-78W	M81757/8-5
MS90365-1	M81757/8-5
MS90365-2	M81757/8-5
M81757/8-2	M81757/8-5 (for Navy use)
M81757/8-3	M81757/8-5 (for Navy use)
M81757/8-2	M81757/8-4

International Standardization. Certain provisions of this specification sheet are the subject of International Standardization Agreement (ASCC AIR STD 12/15 and STANAG 3514L). When revision or cancellation of this specification sheet is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate action through international standardization channels, including department standardization offices, if required.

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